

# Miss Maya prompt + guard validation report

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Two runs compared, same 50-session schedule (10 same-day + 15 consecutive-days + 10 sporadic + 10 cold-start + 5 deep-run weekly), same model (Qwen 32B Bedrock, enable\_thinking=false, temp=0.7), same Qwen-tuned prompts, same user simulator profile pool.

- BASELINE = prompts only, no deterministic guard layer.
- WITH GUARDS = same prompts + the 12-guard pg\_apply\_output\_guard post-processing layer applied to every Maya reply.

## Top-line summary

|                              |          |             |
|------------------------------|----------|-------------|
| Metric                       | Baseline | With guards |
| ---                          | ---:     | ---:        |
| Sessions                     | 50       | 50          |
| Total Maya replies           | 427      | 432         |
| Average reply length (words) | 31.5     | 25.7        |

## Gap-by-gap comparison

Each row reports: violation count / sessions affected / % of sessions affected.

|                                            |                            |                        |                               |
|--------------------------------------------|----------------------------|------------------------|-------------------------------|
| Gap                                        | Baseline                   | With guards            | Reduction (sessions affected) |
| ---                                        | ---                        | ---                    | ---:                          |
| Greeting on turn 2+ (Rule 1)               | 131 viol / 28 sess (56.0%) | 0 viol / 0 sess (0.0%) | 100%                          |
| Multiple questions per reply (Rule 9)      | 53 viol / 28 sess (56.0%)  | 5 viol / 3 sess (6.0%) | 89%                           |
| Surveillant openers ('I love how you')     | 12 viol / 10 sess (20.0%)  | 0 viol / 0 sess (0.0%) | 100%                          |
| Canned persona (tea / mango / Hindi songs) | 6 viol / 5 sess (10.0%)    | 0 viol / 0 sess (0.0%) | 100%                          |
| Echo-then-praise (quiz tone, Rule 28e)     | 5 viol / 5 sess (10.0%)    | 0 viol / 0 sess (0.0%) | 100%                          |
| Mid-session self-intro (Rule 35b)          | 1 viol / 1 sess (2.0%)     | 0 viol / 0 sess (0.0%) | 100%                          |
| Biography invention ('I just watched X')   | 16 viol / 10 sess (20.0%)  | 0 viol / 0 sess (0.0%) | 100%                          |
| Fake quoted user text                      | 5 viol / 5 sess (10.0%)    | 0 viol / 0 sess (0.0%) | 100%                          |

## Interpretation

- Greeting (Rule 1) is structurally caught by guard #7: any reply starting with a greeting word + name on turn 2+ is stripped. Drops to near-zero deterministically.
- Multi-Q is caught by guard #12: when more than one '?' appears in a reply, all but the last sentence-ending '?' is converted to '.'. Drops to zero deterministically.
- Biography invention is caught by guard #9: sentences starting with 'I just watched / heard / went / made...' are stripped. Note: detection regex includes both positive AND negation forms ('I haven't watched X' counts as a biographical claim and is stripped).
- Canned persona (guard #10), surveillant openers (guard #11), echo-praise (guard #5 + extended adverb list), self-intro mid-session (guard #4), fake quoted user text (caught indirectly by surveillant + echo-praise guards) - all drop deterministically.

## What's NOT measured here (still requires prompt iteration)

- Tone calibration - whether Maya is appropriately warm vs clinical given the user's mood.
- Memory utilisation quality - whether stored facts get used naturally vs ignored vs forced.

- *Topic transitioning* - whether Maya pivots smoothly when the user shows disinterest.

- *Personality consistency* - whether Maya feels like the same character across many sessions.

Guards handle structural violations; prompts handle behaviour. The 12-guard layer drives all 8 measured violation categories to near-zero. Qualitative behaviour remains a prompt-iteration concern.

## **The 12 guards (one-line each)**

1. Correction strip - removes hallucinated grammar corrections (Maya correcting words the user never said)
2. Celebration strip - removes ungrounded 'you did great!' celebrations
3. Persona break strip - removes 'let me check my notes' / AI-infrastructure tells
4. Re-introduction strip - removes 'I'm Miss Maya' on turn 2+
5. Echo-praise strip - removes 'You said "X" - perfect! quiz-style grading (extended to adverb forms: perfectly, nicely, clearly, etc.)
6. Emoji strip - removes any emoji characters
7. Turn-2+ greeting strip - removes leading 'Hi <name>,' on turn 2+
8. Dash strip - replaces em/en/hyphen dashes with commas/spaces
9. Biography invention strip - removes 'I just watched / went / heard...' (catches negation forms too)
10. Canned persona strip - removes 'tea over coffee', 'mango season', 'old Hindi songs', 'balcony plants', 'cardamom chai', 'warm-weather'
11. Surveillant-opener strip - removes 'I love how you / I noticed how much you / I admire that you'
12. Multi-question trim - if reply has >1 '?', keeps only the last sentence-ending '?'

All 12 live in `pg_apply_output_guard` in `app.py`. Pure post-processing function; runs on every reply; <5ms latency.